



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# PLASTIC WASTE COLLECTION AND RECYCLING MANAGEMENT SYSTEM

GUIDED BY:-  
Dr. A. JAGATHEESAN

PRESENTED BY:-  
K. PAVAN KUMAR(192524333)  
K. YATHISH  
CHANDRA(192524323)  
T. ADITYA(192565068)

# INTRODUCTION

- Plastic waste has become one of the major environmental challenges due to its increasing use and improper disposal.
- Inefficient waste collection and recycling lead to pollution, harm to wildlife, and health risks.
- A Plastic Waste Collection and Recycling Management System helps in the systematic collection, segregation, and recycling of plastic waste.
- The system improves waste management efficiency by reducing plastic accumulation and promoting resource recovery.
- It supports environmental sustainability by encouraging proper recycling practices and reducing the impact of plastic pollution.

# ABSTRACT

- Plastic waste is increasing every day, and improper disposal causes pollution and harms the environment.
- This project helps collect, track, and manage plastic waste in a simple and organized way.
- The system stores details such as collection location, plastic type, waste weight, and recycling center.
- It uses technologies like AI, IoT, GPS, and data analysis to improve waste collection and recycling.
- The main goal of this project is to reduce plastic pollution, increase recycling, and create a cleaner and healthier environment.

# OBSERVATION

- **Observe the quantity and sources of plastic waste generated** from households, industries, schools, and commercial areas on a daily or weekly basis.
- **Analyze the efficiency of the plastic waste collection system** by monitoring collection frequency, coverage, and proper segregation practices.
- **Examine the effectiveness of the plastic waste segregation process** by identifying recyclable and non-recyclable plastics before recycling.
- **Evaluate the performance of the recycling process** by measuring the amount of plastic successfully processed and converted into reusable products.
- **Assess public participation and awareness** by observing community involvement in plastic waste segregation, recycling programs, and environmental awareness campaigns.

# PROPOSED SOLUTION

- **Develop a smart plastic waste management system** that enables efficient collection, segregation, and recycling of plastic waste using digital technologies.
- **Implement separate collection bins and scheduled waste collection** to ensure proper segregation of recyclable and non-recyclable plastic waste at the source.
- **Use IoT-enabled smart bins and sensors** to monitor bin fill levels in real time and optimize waste collection routes, reducing operational costs and fuel consumption.
- **Integrate a centralized database and monitoring dashboard** to track waste collection, recycling progress, and overall system performance for better decision-making.
- **Promote public awareness and reward-based participation programs** to encourage responsible plastic disposal and increase community involvement in recycling initiatives.

# MODULES

- Module 1: Plastic Waste Collection
- Module 2: Waste Segregation and Recycling
- Module 3: Monitoring and Reporting



# Module 1: Plastic Waste Collection

- Collects plastic waste from households, industries, schools, and public places.
- Records the quantity and type of plastic waste collected.
- Schedules regular waste collection to avoid overflow.
- Ensures proper segregation of plastic waste at the source.
- Transfers collected plastic waste to recycling centers for further processing.

# Module 2: Waste Segregation and Recycling

- Separates collected plastic into recyclable and non-recyclable categories.
- Sorts plastic waste based on its type and quality for efficient recycling.
- Processes recyclable plastic through cleaning, shredding, and recycling.
- Reduces the amount of plastic waste sent to landfills.
- Converts recycled plastic into reusable raw materials or new products.



# Module 3: Monitoring and Reporting

- Monitors the entire waste collection and recycling process in real time.
- Tracks the quantity of plastic waste collected, recycled, and disposed of.
- Generates reports and statistics for performance analysis.
- Identifies areas with low collection efficiency and recycling rates.
- Supports better decision-making through data visualization and system insights.

# TECHNIQUES USED

- **Plastic Waste Segregation** – Separates recyclable and non-recyclable plastic for efficient processing.
- **IoT-Based Smart Bin Monitoring** – Uses sensors to monitor the fill level of waste bins in real time.
- **GPS-Based Waste Collection Tracking** – Optimizes collection routes and monitors waste collection vehicles.
- **Recycling Process Management** – Involves sorting, cleaning, shredding, and reprocessing plastic into reusable materials.
- **Data Monitoring and Reporting** – Collects and analyzes waste management data to generate reports and improve system performance.

# ALGORITHM

## Dijkstra's Algorithm

- **Purpose:** Finds the shortest and most efficient path between waste collection points.
- **Function:** Calculates the minimum travel distance from the waste collection vehicle's starting location to all collection bins.
- **Application:** Helps waste collection vehicles follow the optimal route, reducing travel time and fuel consumption.
- **Advantage:** Improves collection efficiency, minimizes operational costs, and ensures timely waste collection.
- **Outcome:** Enables faster plastic waste collection and supports an efficient recycling management system.

# PROJECT OUTCOMES



# CONCLUSION

- The Plastic Waste Collection and Recycling Management System provides an efficient approach to managing plastic waste.
- It improves waste collection, segregation, and recycling processes, reducing environmental pollution.
- The system promotes sustainable waste management and encourages public participation in recycling activities.
- It helps conserve natural resources by converting plastic waste into reusable materials.
- Overall, the system contributes to a cleaner environment and supports a sustainable future.

# REFERENCES

- **World Health Organization (WHO)** – *Waste Management and Environmental Health*.
- **United Nations Environment Programme (UNEP)** – *Single-Use Plastics: A Roadmap for Sustainability*.
- **Central Pollution Control Board (CPCB), India** – *Plastic Waste Management Guidelines*.
- **Ministry of Environment, Forest and Climate Change (MoEFCC), India** – *Plastic Waste Management Rules, 2016 (and amendments)*.
- **Google Scholar** – Research papers on Plastic Waste Collection and Recycling Management Systems.